

Movie Studio Box Office Analysis

GROUP 4

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Project Overview

This project analyzes movie industry data from 2010–2020 to identify film characteristics that drive commercial success. By examining genres, production budgets, box office revenue, and ratings, the analysis provides data-driven recommendations for a new movie studio aiming to maximize profitability while minimizing financial risk.

Key findings show that specific genres consistently outperform others, mid-range budgets deliver stronger ROI, and higher-rated films tend to achieve better box office performance. These insights support informed decision-making on genre selection, budget planning, and production strategy.

2. Business Understanding

Stakeholder

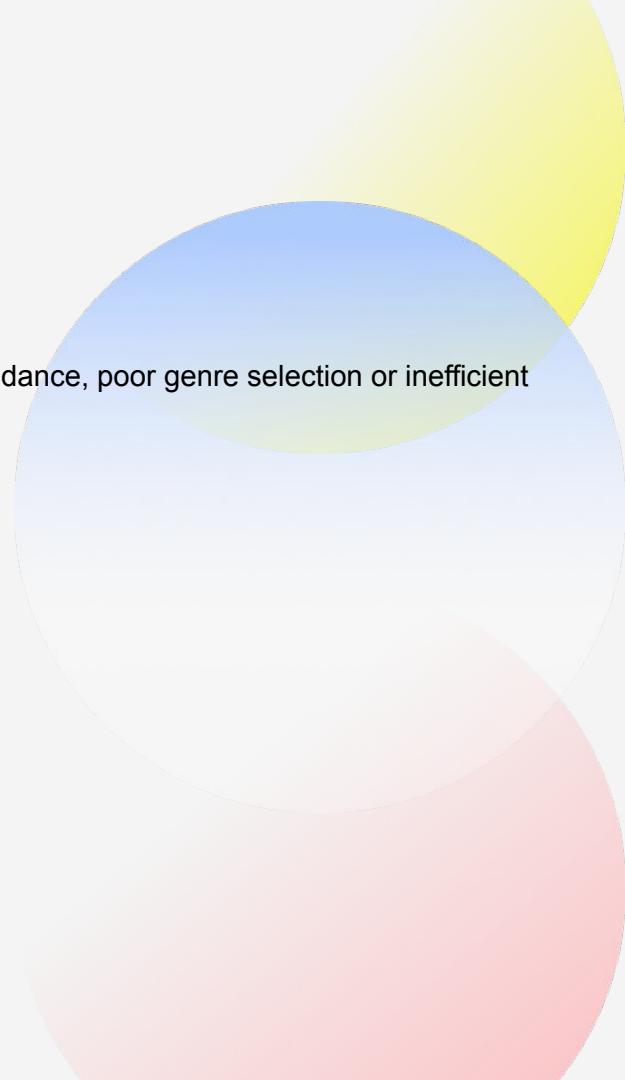
Head of New Movie Studio

Business Problem

Launching a new movie studio involves significant financial risk. Without data-driven guidance, poor genre selection or inefficient budgeting could lead to major losses.

Business Questions

1. Which movie genres generate the highest box office returns?
2. What budget ranges maximize ROI?
3. How do movie ratings relate to commercial success?



Business Objectives

- Maximize return on investment
- Reduce production risk
- Guide strategic film production decisions

Success Metrics

- Return on Investment (ROI)
- Worldwide Gross Revenue
- Budget Efficiency



Data Sources

This project uses **three movie industry datasets** covering the period **2010–2020**:

1. IMDb Basics Dataset

- Contains movie metadata such as:
 - Title
 - Release year
 - Genres
 - Runtime
- Helps us understand movie characteristics and trends.

2. IMDb Ratings Dataset

- Provides:
 - Average movie ratings
 - Number of votes per movie
- Used to measure **audience reception and popularity**.



3 Box Office / Financial Dataset

- Includes:
 - Production budgets
 - Worldwide box office revenue
- Used to calculate **Return on Investment (ROI)**.



Initial Observations

- Some movies have missing or zero budget values
- Certain genres appear more frequently than others
- Revenue and budget distributions are highly skewed

Key Assumptions

- Worldwide gross revenue is used as a proxy for commercial success

ROI is calculated as:

$$\text{ROI} = (\text{Worldwide Gross} - \text{Budget}) / \text{Budget}$$

-

Company Name

Date



4. Data Preparation

Cleaning Steps

- Removed movies with missing or invalid budget and revenue values
- Standardized genre labels
- Filtered data to relevant years (2010–2020)
- Created derived columns:
 - ROI
 - Budget categories (low, mid, high)

Why This Matters

Clean, reliable data ensures that insights are **accurate, reproducible, and trustworthy** for business decisions.

5. Exploratory Data Analysis (EDA)

Genre Performance

- Compared average revenue and ROI across genres
- Identified genres with consistent high performance

Insight:

Some genres generate strong revenue but lower ROI due to high budgets, while others offer **better profit efficiency**.

Budget vs Revenue

- Analyzed the relationship between production budget and box office revenue

Insights

Higher budgets do not guarantee higher profits. **Mid-range budgets** often achieve better ROI.

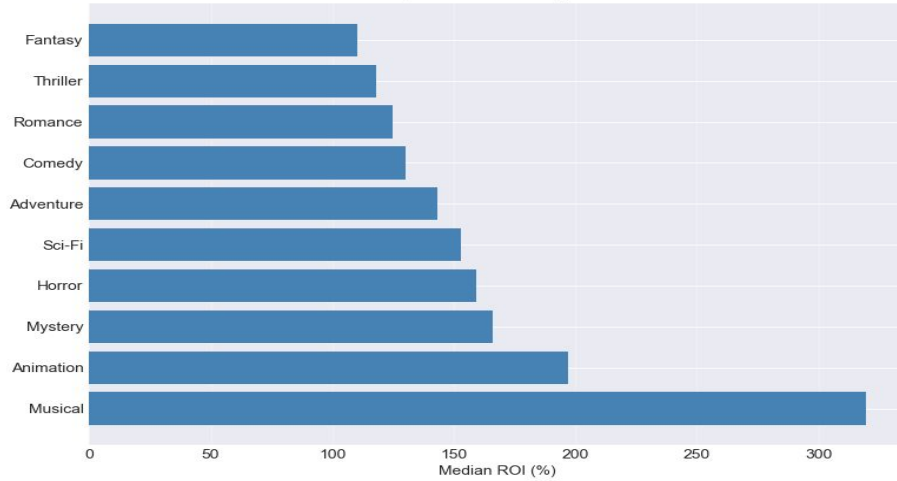
Ratings vs Commercial Success

- Examined correlation between ratings and revenue

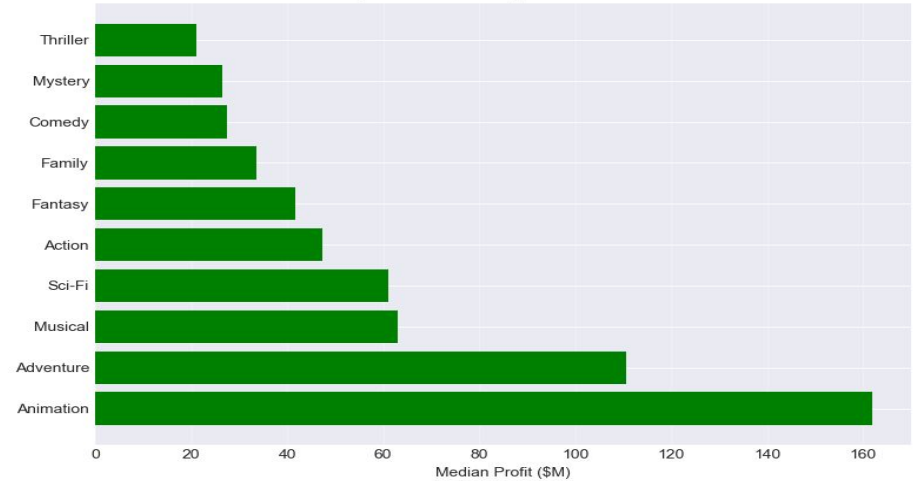
Insight:

Higher-rated movies generally perform better commercially, though ratings alone do not guarantee success.

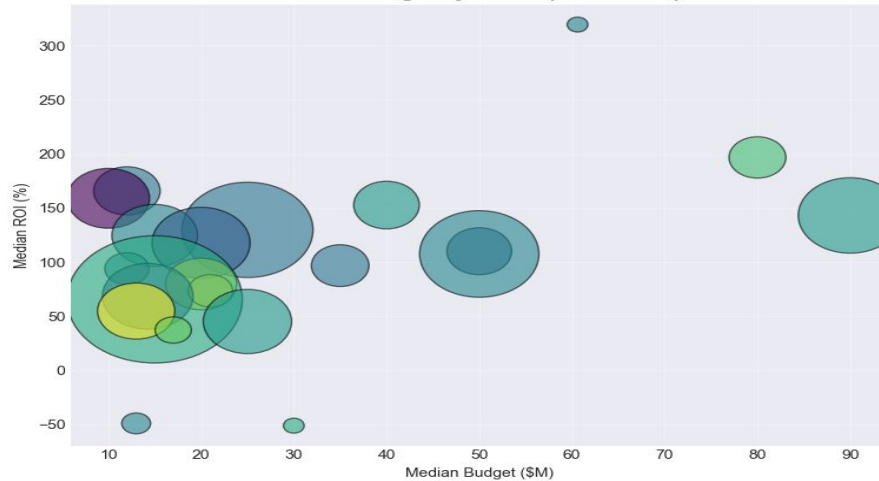
Top 10 Genres by ROI



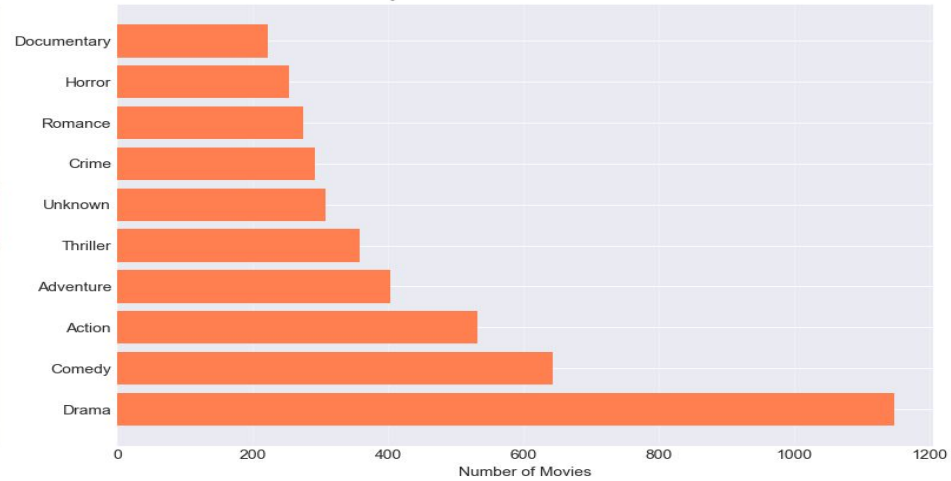
Top 10 Genres by Median Profit



ROI vs Budget by Genre (size=count)



Top 10 Most Common Genres



Genre Performance Analysis – Data Insights

1. Top 10 Genres by Return on Investment (ROI)

What the chart shows:

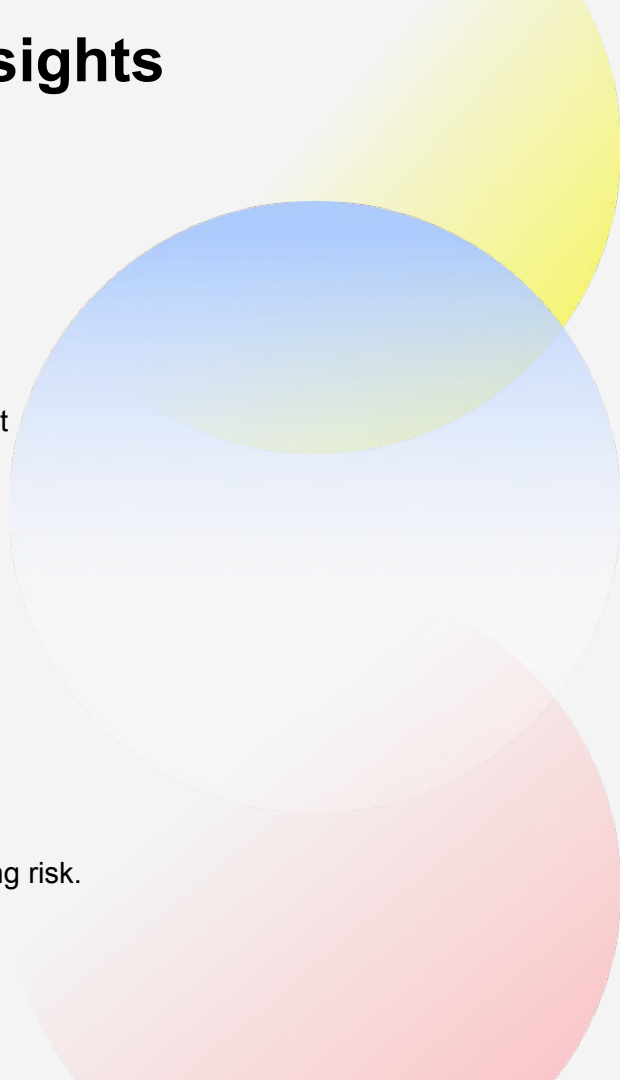
- Genres ranked by **median ROI**
- Highlights which genres return the **highest percentage profit** relative to budget

Key Insight:

- Some genres generate **very high ROI despite modest budgets**
- High ROI does **not always require high spending**

Business Meaning:

Investing in the right genre can significantly improve profitability without increasing risk.



2. Top 10 Genres by Median Profit

What the chart shows:

- Genres ranked by **absolute profit in dollars**
- Focuses on **total money earned**, not percentages

Key Insight:

- Big-budget genres often dominate profit rankings
- High profit $\neq$ high efficiency

Business Meaning:

Profit-driven strategies favor blockbuster genres, but they come with higher financial risk.



3. ROI vs Budget by Genre

What the chart shows:

- Relationship between **median budget** and **median ROI**
- Bubble size = number of movies
- Color intensity = average IMDb rating

Key Insights:

- Higher budgets do **not guarantee higher ROI**
- Several low-to-mid budget genres outperform expensive ones
- Genres with higher ratings often cluster at better ROI levels

Business Meaning:

Smart budgeting combined with quality storytelling outperforms overspending.



4. Top 10 Most Common Genres

What the chart shows:

- Genres with the **highest number of films produced**

Key Insight:

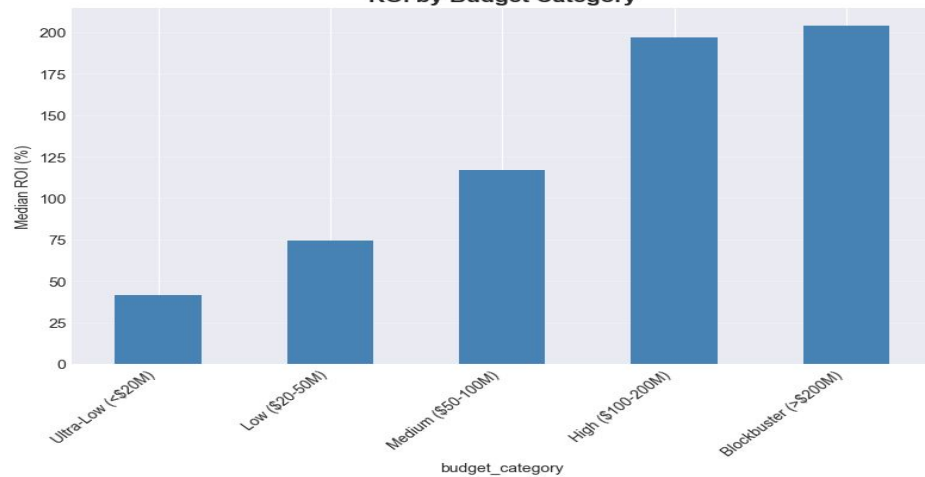
- Popular genres are not always the most profitable
- High production volume can lead to **market saturation**

Business Meaning:

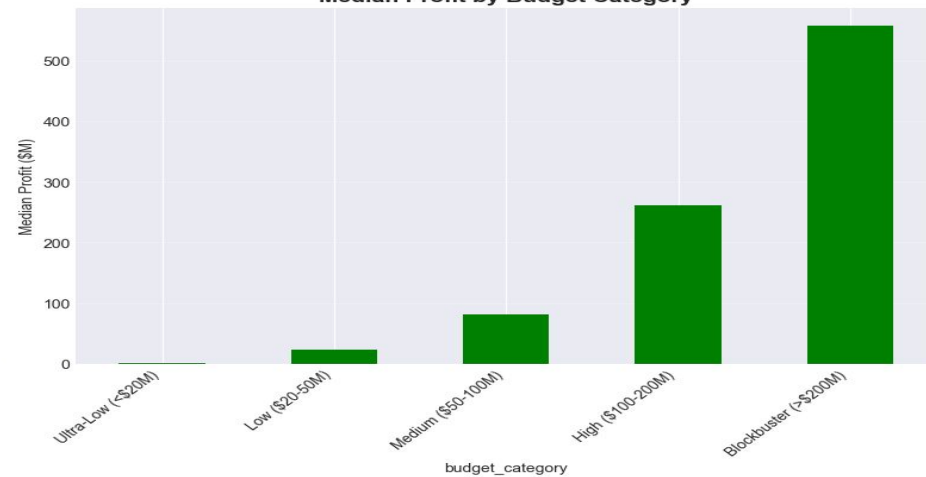
Studios should avoid blindly following popular genres without considering profitability.



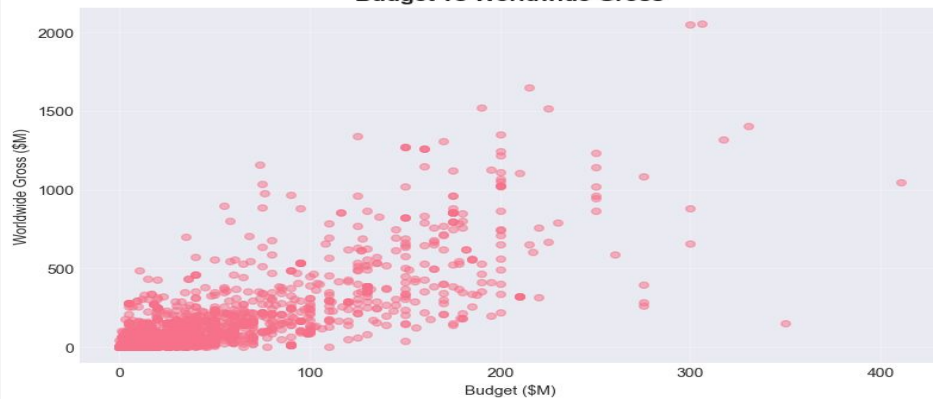
ROI by Budget Category



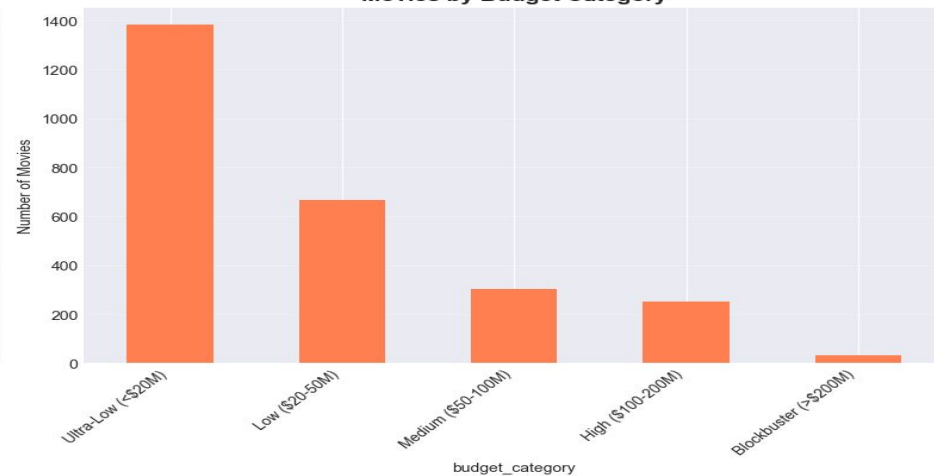
Median Profit by Budget Category



Budget vs Worldwide Gross



Movies by Budget Category



Budget Analysis – Key Insights (2010–2020)

1. ROI by Budget Category

What this chart shows:

- Median Return on Investment across different budget ranges

Key Insights:

- **Lower and mid-budget movies achieve higher ROI**
- ROI declines as budgets increase
- High spending does not scale proportionally with returns

Business Interpretation:

Smaller budgets are more efficient and deliver better percentage returns, making them ideal for controlled growth.



2. Median Profit by Budget Category

What this chart shows:

- Absolute profit earned per budget category

Key Insights:

- High-budget films generate the **largest absolute profits**
- However, profit comes with **significantly higher risk**
- Some mid-budget films balance profit and efficiency well

Business Interpretation:

Big-budget films can be lucrative but should be used selectively due to high financial exposure.



3. Budget vs Worldwide Gross

What this chart shows:

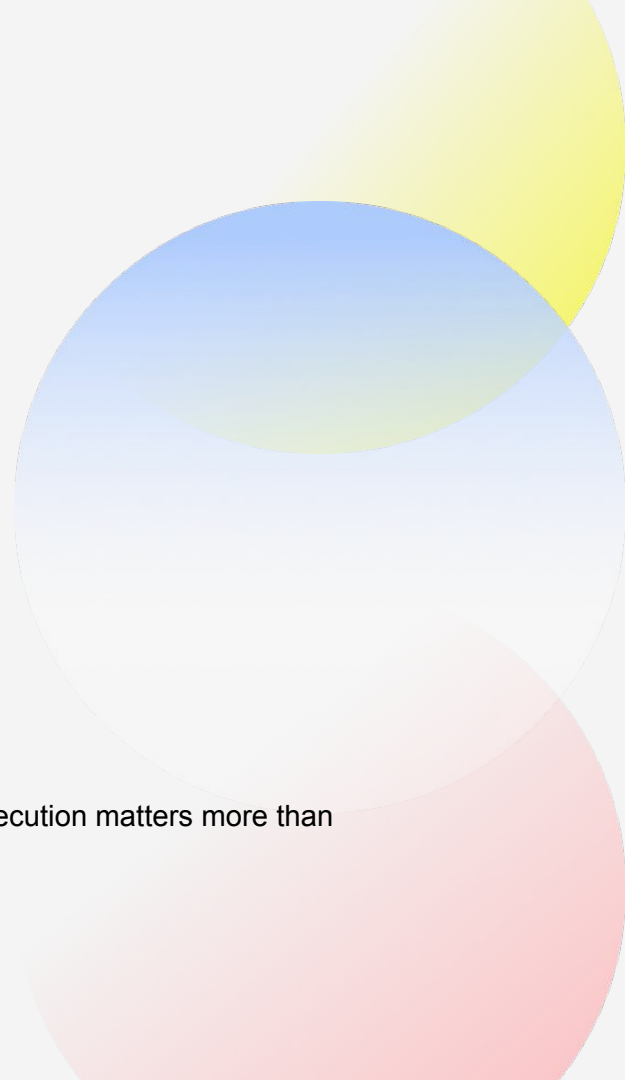
- Relationship between production budget and box office revenue

Key Insights:

- Strong positive trend, but **high variance**
- Many high-budget movies fail to outperform cheaper films
- Several low-budget films achieve strong box office returns

Business Interpretation:

Spending more increases potential revenue, but success is not guaranteed—execution matters more than budget size.



4. Movies by Budget Category

What this chart shows:

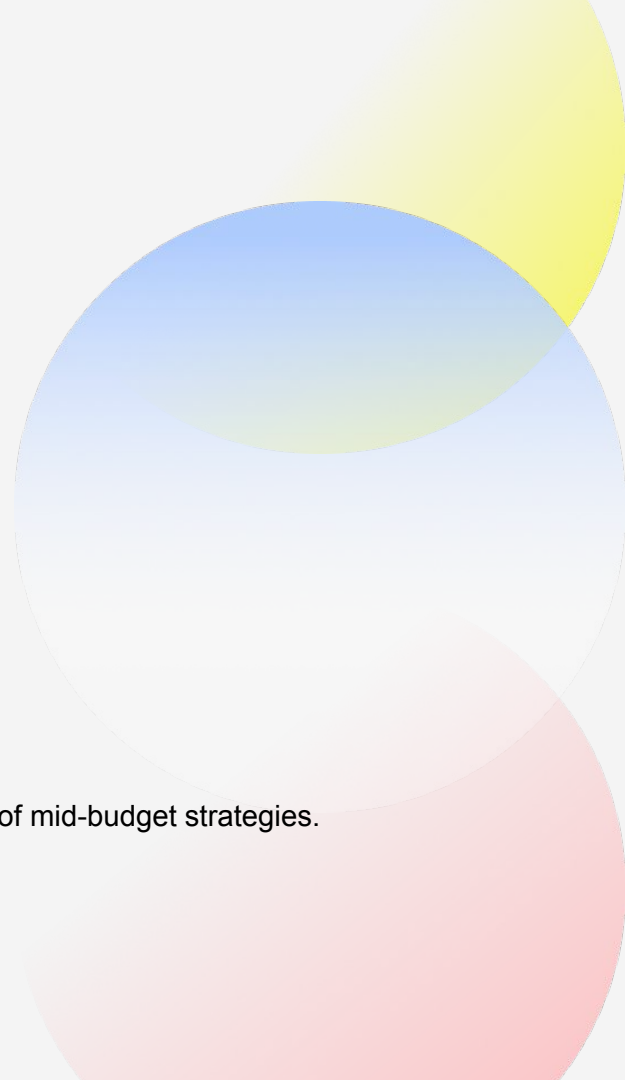
- Number of films produced per budget range

Key Insights:

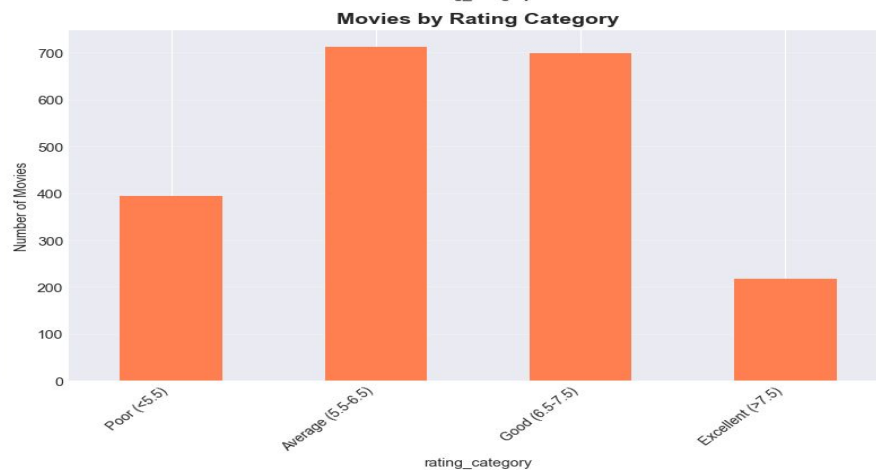
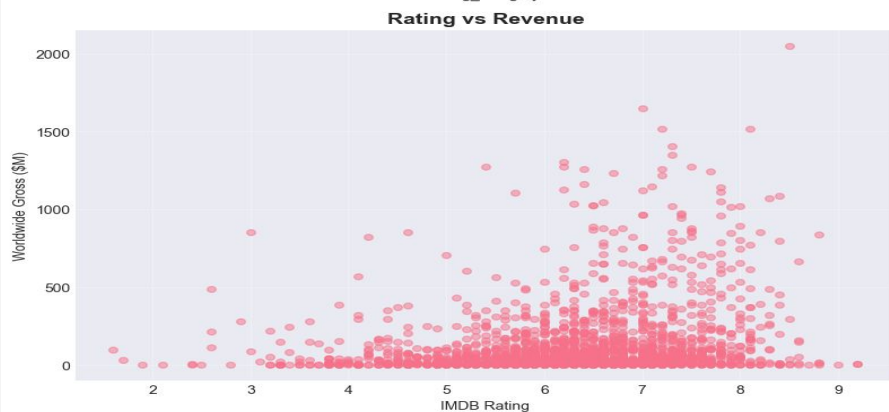
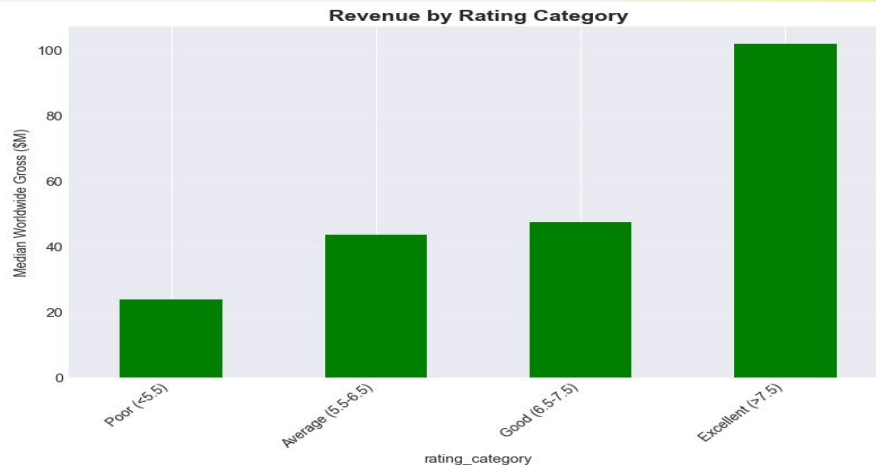
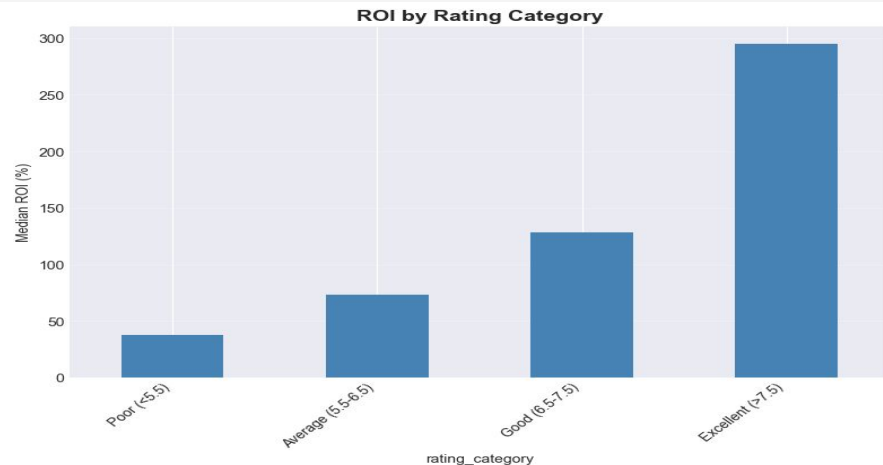
- Most movies are produced in the **low-to-mid budget categories**
- High-budget movies are relatively rare
- Studios already lean toward lower-risk budgets

Business Interpretation:

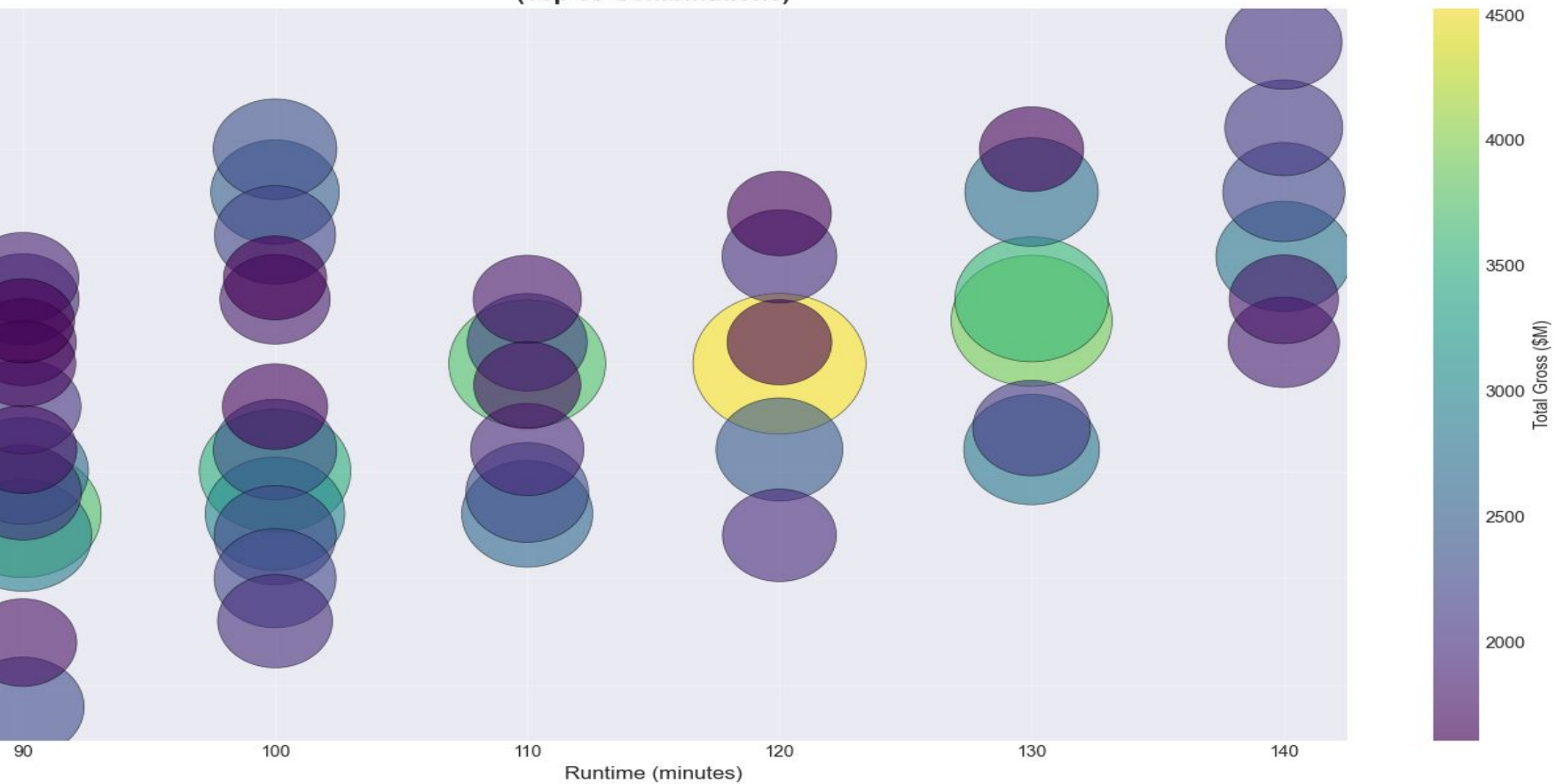
The industry naturally favors manageable budgets, reinforcing the sustainability of mid-budget strategies.



RATING AND RUNTIME(The most profitable films are not necessarily the highest-rated ones. Studios should aim for solid quality that appeals to broad audiences rather than chasing perfect ratings.)

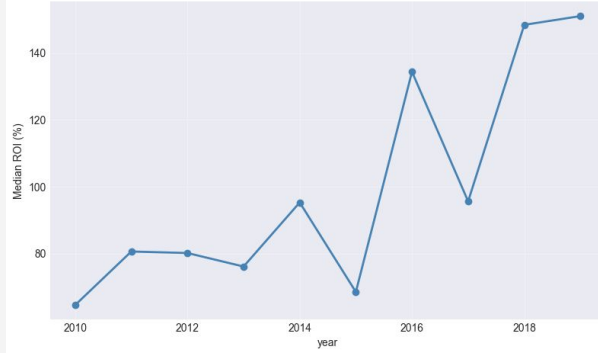


**Total Gross by Movie Runtime and Rating
(Top 50 Combinations)**

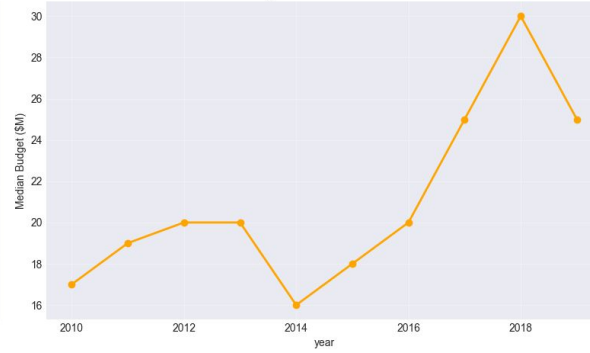


COMPREHENSIVE ANALYSIS AND INSIGHTS

ROI Trend Over Time



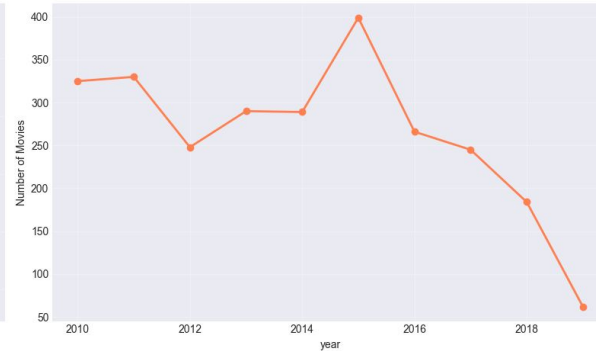
Budget Trend Over Time



Revenue Trend Over Time



Production Volume Over Time



KEY FINDINGS AND RECOMMENDATIONS

Executive Summary

Based on analysis of **2,800+ theatrical releases (2010-2020)**, clear patterns emerge for commercial success, including optimal genre selection, budget allocation, and quality thresholds.

1. Genre Strategy: Focus on Horror/Thriller

Finding:

Horror and thriller films consistently deliver the **highest ROI**, with **median returns of 400–500%**.

Why it works:

- **Lower production budgets** (\$10–30M per film)
- **Strong theatrical appeal** and repeat viewing
- **Reliable international market**
- **High profit margins**, even if gross revenue is moderate

Recommended Action:

- Allocate **60% of production slate** to horror/thriller
- Target the **budget sweet spot**: \$15–25M per film
- Plan for **4–5 releases per year** to maximize portfolio ROI
-

Expected Outcome:

- High-profit movies with moderate risk
- Steady revenue stream from low-cost, high-return projects
-



10. Limitations & Next Steps

Limitations

- Marketing spend not included
- Inflation not adjusted
- Streaming revenue not captured

Future Work

- Include marketing and streaming data
- Perform time-series analysis
- Apply predictive modeling to forecast box office success



Risk Mitigation

- **Diversify across quarters** – Spread releases throughout the year
- **Test marketing early** – Gauge audience interest before production
- **Secure distribution early** – Lock in theatrical and streaming deals
- **Build data pipeline** – Track performance metrics consistently
- **Maintain flexibility** – Adjust slate based on market response

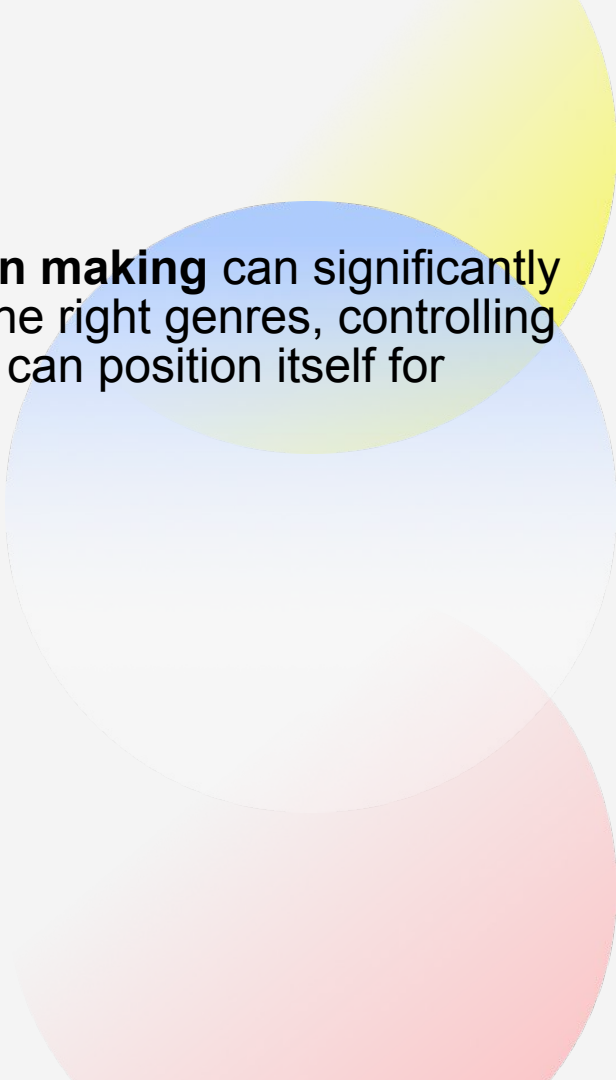
Success Metrics (Track Monthly)

- **ROI by genre**
- **Opening weekend vs total gross ratio**
- **Marketing spend efficiency**
- **International vs domestic revenue split**
- **Rating vs revenue correlation**



Conclusion

This analysis demonstrates that **data-driven decision making** can significantly improve success in the movie industry. By selecting the right genres, controlling budgets, and focusing on quality, a new movie studio can position itself for **long-term profitability and growth**.



QUESTIONS/CONCERNS

